

1 Strategic Comparison to Fixed-Bet and No-Limit Continuous Poker

LCP interpolates between Fixed-Bet Continuous Poker (FBCP) and No-Limit Continuous Poker (NLCP). We now make this precise by showing that the LCP strategies converge to those of FBCP and NLCP as the betting limits approach appropriate boundary values.

1.1 Notation

We use the following notation for strategy functions across variants:

- $S_{FB}(x, B), C_{FB}(s, B)$: Bettor and caller strategies in FBCP with fixed bet B
- $S_{NL}(x), C_{NL}(s)$: Bettor and caller strategies in NLCP
- $S_{LCP}(x, L, U), C_{LCP}(s, L, U)$: Bettor and caller strategies in LCP

The FBCP strategies are:

$$S_{FB}(x, B) = \begin{cases} B & x < \frac{B}{(1+2B)(2+B)} \text{ (bluff)} \\ 0 & \frac{B}{(1+2B)(2+B)} < x < \frac{1+4B+2B^2}{(1+2B)(2+B)} \text{ (check)} \\ B & x > \frac{1+4B+2B^2}{(1+2B)(2+B)} \text{ (value)} \end{cases}$$

$$C_{FB}(B) = \frac{B(3+2B)}{(1+2B)(2+B)}.$$

The NLCP strategies are:

$$S_{NL}(x) = \begin{cases} b_{NL}^{-1}(x) & x < \frac{1}{7} \text{ (bluff)} \\ 0 & \frac{1}{7} < x < \frac{4}{7} \text{ (check)} \\ v_{NL}^{-1}(x) & x > \frac{4}{7} \text{ (value)} \end{cases}$$

$$C_{NL}(s) = 1 - \frac{6}{7(s+1)},$$

where $v_{NL}(s) = 1 - \frac{3}{7(s+1)^2}$ and $b_{NL}(s) = \frac{3s+1}{7(s+1)^3}$.

1.2 Strategic Convergence

The LCP strategy functions are rational in L and U , so convergence follows from continuity by direct substitution of limit values.

Remark (Convergence to FBCP). As $L, U \rightarrow B$ for any fixed $B > 0$:

$$S_{LCP}(x, L, U) \rightarrow S_{FB}(x, B), \quad C_{LCP}(s, L, U) \rightarrow C_{FB}(s, B).$$

At $L = U = B$, the intermediate bet-size regions collapse ($x_0 = x_1$ and $x_4 = x_5$), leaving only the fixed bet B as an option. The thresholds reduce to:

$$x_2 = \frac{B}{(1 + 2B)(2 + B)}, \quad x_3 = \frac{2B^2 + 4B + 1}{(1 + 2B)(2 + B)},$$

which match the FBCP bluff/check and check/value boundaries.

Remark (Convergence to NLCP). As $L \rightarrow 0$ and $U \rightarrow \infty$:

$$S_{LCP}(x, L, U) \rightarrow S_{NL}(x), \quad C_{LCP}(s, L, U) \rightarrow C_{NL}(s).$$

At these limits, the boundary regions collapse ($x_0 \rightarrow 0$, $x_5 \rightarrow 1$), and the thresholds become:

$$x_1 = x_2 \rightarrow \frac{1}{7}, \quad x_3 = x_4 \rightarrow \frac{4}{7},$$

matching the NLCP bluff/check and check/value boundaries. The bet-size functions converge to

$$b(s) \rightarrow \frac{3s + 1}{7(s + 1)^3}, \quad v(s) \rightarrow 1 - \frac{3}{7(s + 1)^2},$$

which are exactly $b_{NL}(s)$ and $v_{NL}(s)$. The calling threshold converges to $c(s) \rightarrow 1 - \frac{6}{7(1+s)} = C_{NL}(s)$.

These convergence results confirm that Nash equilibria of LCP smooth interpolate between optimal strategies in the two existing variants.